Benchmarking TCR–pMHC structure prediction without multiple sequence alignments: a protein language model matches AlphaFold 3, and their disagreement predicts accuracy

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Abstract

T cell receptor (TCR) recognition of peptide–MHC (pMHC) is the central specificity event of cellular immunity, and structure prediction is now the main route to modelling it at scale. Published benchmarks agree that multiple-sequence-alignment (MSA)-based predictors, and AlphaFold 3 in particular, lead the field, and that protein-language-model (PLM) predictors trail them. We revisited that conclusion on a benchmark of 136 TCR–pMHC entries released after the 30 September 2021 training cutoff shared by every method tested, of which 126 (111 class I, 15 class II) carry a scoreable experimental reference. All methods received byte-identical truncated inputs and a single fixed chain convention, and every model was scored with an explicit identity chain mapping against the same experimental references, cropped from the deposited coordinates to the input sequence and verified to draw all chains from a single author-defined assembly. AlphaFold 3 (median Global DockQ 0.751) and ESMFold2, a single-sequence diffusion predictor conditioned on a 6-billion-parameter language model rather than on an MSA (0.736), were statistically indistinguishable (paired Wilcoxon p = 0.68); both exceeded Protenix (0.701, p = 0.045 and p = 0.0009) and the TCR-specialised TCRmodel2 (0.708 on the 111 class I targets) was not separable from either. Equal accuracy did not mean equivalent behaviour. ESMFold2's ipTM tracked accuracy far more tightly than AlphaFold 3's (Pearson r = 0.79 vs 0.66); AlphaFold 3 was more sensitive to reference resolution; and the two models placed the TCR more than 90° apart on 10 of 126 complexes while agreeing on the pMHC to within 0.44 Å. The Cα RMSD between the two predictions — a quantity computable without any experimental structure — predicted the worse of the two DockQ scores better than either model's own confidence (Spearman $\rho = -0.884$ vs +0.60 and +0.64), and the two methods together reached medium-or-better accuracy on 119 of 126 complexes against 107 and 115 individually. Instrumenting both samplers showed the equal accuracy is reached by different routes: AlphaFold 3 is still 44 Å from its answer at the midpoint of denoising and converges only in the final fifth of its schedule, whereas ESMFold2 is within 6 Å at its midpoint. Exploiting that, cutting AlphaFold 3 from 200 to 160 denoising steps and from 5 diffusion samples to 1 gave a 1.80× speed-up on 30 complexes at a cost of −0.010 Global DockQ (95% CI −0.032 to +0.007, p = 0.33). We conclude that MSAs are not required for state-of-the-art TCR–pMHC modelling, that cross-architecture agreement is a strong reference-free quality signal, and that the default AlphaFold 3 sampler is substantially over-provisioned for complexes of this size.

1 Introduction

The $\alpha\beta$ T cell receptor (TCR) recognises a short peptide presented in the groove of a major histocompatibility complex (MHC) protein: a groove closed at both ends in class I, which therefore binds peptides of restricted length [1], and open at both ends in class II, which admits longer ones [2]. The first ternary structures established that the receptor sits on this composite peptide–MHC surface in a conserved, roughly diagonal polarity, with its α and β variable domains over the two MHC helices [3,4], an arrangement that has held across the complexes solved since [5] and that is thought to be biased by germline-encoded contacts between the CDR1 and CDR2 loops and the helices themselves [6]. Specificity for the peptide is carried mainly by the hypervariable CDR3 loops, which are generated by V(D)J recombination and junctional diversity and lie over the centre of the groove [7,8]. A structural model of a TCR–pMHC complex must therefore get two distinct things right — the docking geometry of the receptor on the platform, and the conformations of the CDR loops that contact the peptide — and, as this work shows, they are not equally hard.

The scale of the repertoire is what makes computational modelling necessary. A single individual carries on the order of a million or more distinct $\alpha\beta$ clonotypes [9], each of which must be substantially cross-reactive for the repertoire to cover the space of possible antigens: a single TCR can recognise more than a million different peptides [10,11], and the peptide preferences of one receptor span a degenerate sequence space rather than a single epitope [12]. Sequence-based methods can group TCRs of shared specificity where enough labelled examples exist [13,14], but they generalise poorly to unseen epitopes and recover no structural basis for the recognition they predict. The experimental record, meanwhile, grows slowly: the structural databases that curate TCR–pMHC complexes hold only a few hundred ternary structures [15–17]. With TCR-based therapeutics now reaching the clinic [18], computational modelling has become the practical route to structural hypotheses at scale, whether for epitope-specificity prediction, TCR engineering, or immunotherapeutic design.

Deep-learning structure prediction transformed this problem twice. AlphaFold 2 made accurate single-chain and many multi-chain predictions routine [19], though early benchmarks identified adaptive immune recognition — antibody–antigen and TCR–pMHC complexes in particular — as a systematic failure mode [20]. AlphaFold 3 replaced the Structure Module with a diffusion head and reported substantially improved interface accuracy across biomolecular space [21], and TCR-specific adaptations such as TCRdock [22] and TCRmodel2 [23] narrowed the gap for this class of complex specifically. In parallel, a third line of work made structure prediction possible without an MSA at all: ESMFold demonstrated that a sufficiently large language model encodes atomic-level structure in its representations [24], trading evolutionary coupling signal for a learned prior and an order-of-magnitude speed-up. That exchange has so far been understood as a trade — single-sequence predictors bought speed and paid accuracy, and on immune complexes specifically they paid a great deal of it. Whether the alignment is still doing indispensable work, or has become a component that a large enough language model can replace outright, is the question this benchmark was built to answer.

Recent TCR–pMHC benchmarks have converged on a consistent ranking. Lu et al. evaluated thirteen methods on 70 post-cutoff complexes and concluded that "MSA-based methods, particularly AlphaFold3, consistently outperform all alternatives", with most PLM-based models underperforming except a PLM variant built on an AlphaFold-3-style architecture [25]. Ascunce-París et al. reached the same conclusion on 265 class I complexes, reporting AlphaFold 3 as the best generaliser to post-training structures [26]. Shi et al. found that TCR-specific tools derived from AlphaFold 2 gain interface accuracy at the cost of framework accuracy, and that CDR3 loops, docking orientation and class II MHC–peptide interfaces remain difficult for every method [27]. Yin et al. observed complementary behaviour between methods and pooling gains for antibody–peptide complexes [28]. Together these establish the field's working assumption: evolutionary information is what makes TCR–pMHC prediction work.

Two considerations motivate re-examining that conclusion. First, that consensus was established against a specific and now superseded generation of single-sequence predictor: the PLM entries in those comparisons are predominantly ESMFold-generation models whose structure module is a direct AlphaFold-2 descendant, and which handle multi-chain inputs by concatenating the chains through a poly-glycine linker rather than modelling them as a complex at all. A current-generation language-model predictor that pairs a much larger language model with a diffusion structure head is a substantially different architecture, and the published ranking does not speak to it. Second, a benchmark that ranks methods addresses only part of the question. Where two architectures that share little on the conditioning side — one using a paired MSA and structural templates, the other only the query sequence — achieve comparable accuracy, the more informative questions concern the route each takes to that accuracy and whether the disagreements between them carry usable information.

Four methods are scored on every measure below, but two of them carry the comparisons that follow, and it is worth saying why at the outset. AlphaFold 3 and ESMFold2 are the two most accurate on this benchmark (median Global DockQ 0.751 and 0.736, not separable from each other at $p = 0.68$, and both separating from Protenix), and they are also the pair that differs most in what a prediction is conditioned on: AlphaFold 3 runs a diffusion head on top of a multiple sequence alignment and structural templates, while ESMFold2 runs a diffusion head of the same generative family on the output of a 6-billion-parameter protein language model, with no alignment and no template. Their disagreement therefore isolates the contribution of the conditioning rather than of the generative architecture. The other two methods are informative but not suited to that contrast: Protenix is an open reproduction of the AlphaFold 3 architecture, so it shares AlphaFold 3's conditioning as well as its generative head, and TCRmodel2 is an AlphaFold 2 derivative restricted to class I here. Analyses that need a matched pair — the structural divergence of Section 3.9 and the sampler trajectories of Section 3.10 — therefore use AlphaFold 3 and ESMFold2.

Here we assemble a benchmark of 136 TCR–pMHC entries released after 30 September 2021, the training cutoff shared by AlphaFold 3, Protenix and ESMFold2, of which 126 carry a scoreable experimental reference. Every method receives identical truncated inputs under one chain convention, and every prediction is scored against a native cropped from the deposited coordinates and audited to come from a single author-defined assembly. We report interface accuracy (DockQ), TCR-specific accuracy (MHC-aligned weighted all-CDR Cα RMSD and framework-aligned CDR3 all-atom RMSD), confidence calibration, sensitivity to reference resolution and to training-set similarity, the structural anatomy of the disagreement between the two leading models, the internal trajectories of both diffusion samplers, and the wall-clock cost of the default AlphaFold 3 sampler settings.

2 Materials and methods

2.1 Benchmark set construction

We constructed a dataset of TCR-pMHC Class I and Class II complexes from the TCR3d database [15,17], which curates all known TCR complex structures. Restricting to $\alpha\beta$ TCR complexes with an initial RCSB Protein Data Bank [29] release date after 30 September 2021 gave 427 class I and 72 class II candidates. Of these, 136 were complete TCR–peptide–MHC complexes containing all of an MHC peptide-binding platform, a peptide, and paired TCR α and β variable domains (114 class I, 22 class II); the remainder were unliganded TCRs, pMHC-only entries, TCR-mimic antibody complexes, or otherwise incomplete triads.

All 136 entries were reviewed against their deposited coordinates and primary publications, and ten were excluded from scoring. Five (8TRL, 8TRQ, 8TRR, 9NIG, 9WNT) contain modified peptide residues that a standard FASTA sequence cannot represent, and because the modification is in every case part of the interaction being modelled, predicting these complexes from sequence alone is not well posed. Four (8VD0, 8VQ8, 9AUD, 9C3E) are engineered constructs in which the peptide is covalently fused to the MHC through a linker; modelling their components as separate chains would alter the connectivity of the experiment. The last, 9J4S, has no physical TCR–pMHC interface in any author-defined assembly of the deposited structure and so cannot be scored at all. This leaves 126 scoreable complexes, 111 class I and 15 class II.

No redundancy filter or resolution cut was applied to the primary set. TCR variable-domain identity is a poor novelty axis — germline V-gene framework dominates a ~110-residue V domain — and filtering on it discards most of the set while retaining an idiosyncratic remainder (Section 3.8). Instead, similarity to the pre-cutoff structural record is measured explicitly and treated as a covariate.

2.2 Domain truncation

Each complex was truncated to the domains that constitute the variable regions of the TCR–pMHC interface (Figure 1) — the same reduction to peptide-binding platform, peptide and TCR variable domains used by TCRdock [22] and by recent benchmarks of this class of complex [25]. For class I, β2-microglobulin was removed and the MHC heavy chain was cut to its $\alpha 1 \alpha 2$ peptide-binding platform (median 181 residues, range 161–181); for class II, the α1 and β1 domains were retained (81–85 and 89–93 residues). TCR chains were cut to their variable domains (α 101–114 residues, β 105–116). Prediction inputs used the complete deposited polymer sequences over those spans.

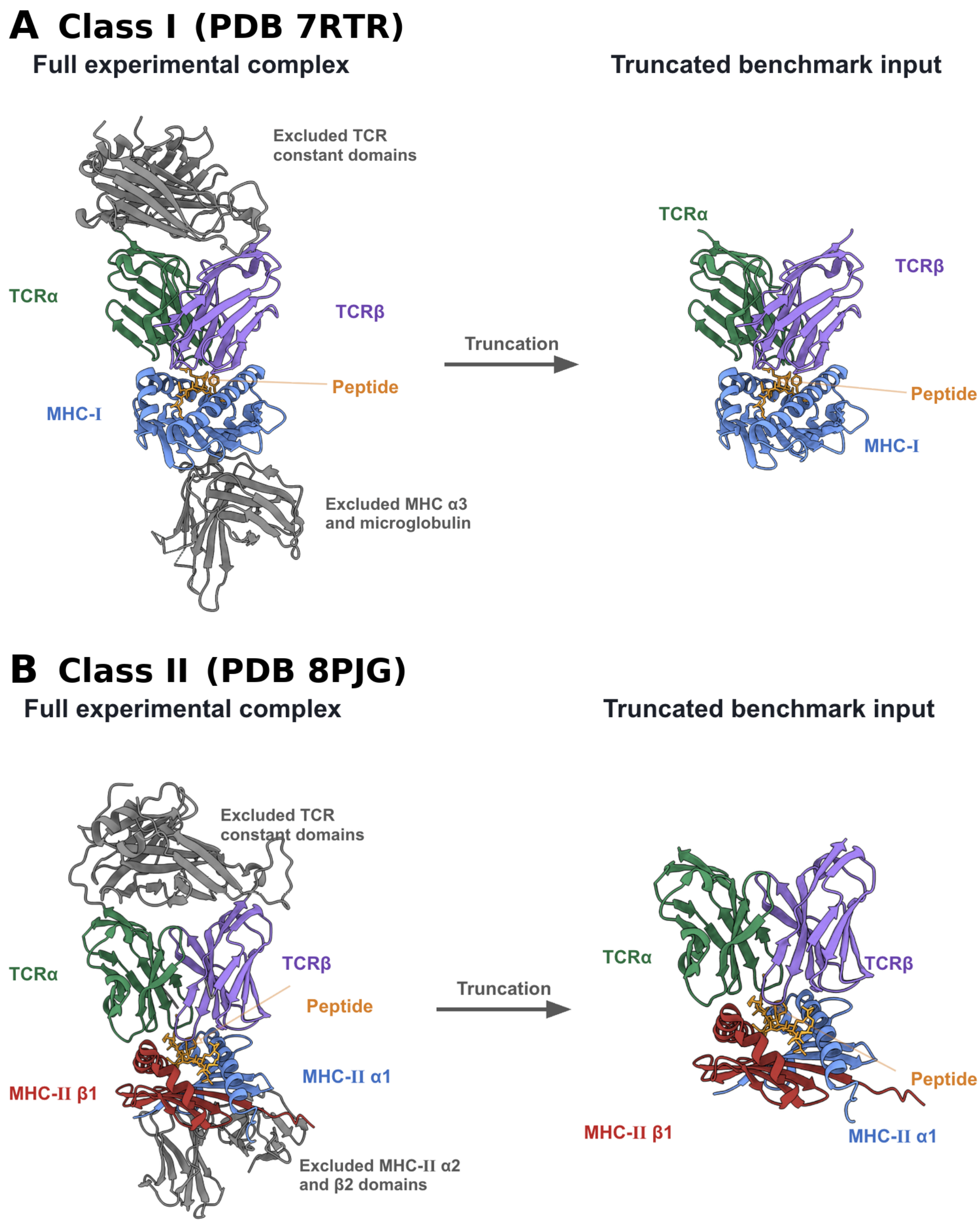

Figure 1: **Domain truncation applied to the benchmark complexes.** (**A**) Class I, PDB 7RTR. The TCR constant domains, the MHC $\alpha3$ domain and $\beta2$-microglobulin are removed, leaving the MHC-I $\alpha1\alpha2$ peptide-binding platform, the peptide and the paired TCR variable domains. (**B**) Class II, PDB 8PJG. The TCR constant domains and the MHC-II $\alpha2$ and $\beta2$ domains are removed, leaving the $\alpha1$ and $\beta1$ peptide-binding domains, the peptide and the paired TCR variable domains. Retained chains are coloured by role; removed regions are grey.

2.3 Structure prediction methods

Four methods were run on identical inputs.

AlphaFold 3 [21] predictions were obtained through the AlphaFold Server, which supplies MSAs and structural templates from its own pipeline and returns five ranked diffusion samples; the top-ranked model by ranking score was scored. The server enforces a 30 September 2021 training cutoff.

Protenix [30], an open reproduction of the AlphaFold 3 architecture, was run through its public server with the same inputs, and the top-ranked model scored.

ESMFold2 (the `biohub/ESMFold2` checkpoint, conditioned on the `biohub/ESMC-6B` language model) [31] was run through the ESM SDK API as model `esmfold2-fast-2026-05` with the SDK defaults of 10 refinement loops and 100 sampling steps. ESMFold2 is a diffusion structure predictor in the same sense as AlphaFold 3, but it takes no MSA and no structural template: its only input is the query sequence, embedded by a 6-billion-parameter protein language model. This is the architectural contrast on which the present comparison rests, and it is a strict one, since the two methods share the diffusion generative head and little else on the conditioning side.

TCRmodel2 [23] was installed locally from the Pierce laboratory repository (AlphaFold 2.3 base, `alphafold_params_2022-12-06`) with `--max_template_date=2021-09-30` applied explicitly to both the TCR and the pMHC template searches; the default is effectively unrestricted, so this setting is required for a valid post-cutoff comparison. TCRmodel2 accepts a fixed peptide length for class II and was therefore run on class I only, giving 111 predictions.

Mean end-to-end runtimes were 5.2 s per complex for the ESMFold2 API (median, IQR 4.9–6.4 s) and 13.4 s per complex for a local AlphaFold 3 run at default sampler settings on one RTX PRO 6000 (Section 2.8).

2.4 Interface accuracy

Interface accuracy was measured with DockQ v2.1.3 [32,33], run once per complex. The primary reported quantity is DockQ v2's Global DockQ, its interface-size weighted score over the whole complex; per-pairwise-interface DockQ values are also retained, giving 6 interfaces per class I complex and 10 per class II complex. Standard CAPRI thresholds are used throughout (< 0.23 incorrect, 0.23–0.49 acceptable, 0.49–0.80 medium, ≥ 0.80 high) [32,34].

2.5 TCR-specific RMSD metrics

Two TCR-specific metrics were computed on the same top-ranked predictions the DockQ workflow scores; predictions were never selected using a native RMSD. TCRα and TCRβ chains were annotated once per biological sequence with ANARCI [35] under IMGT numbering [36], and the annotation propagated to prediction and native coordinates by pairwise sequence alignment rather than by PDB residue numbering; CDRs are the inclusive IMGT intervals 27–38 (CDR1), 56–65 (CDR2), 81–86 (CDR2.5) and 104–118 (CDR3) of Woods et al. [37], and "all CDRs" means all four regions on both chains, eight in total. The first, the MHC-aligned weighted all-CDR Cα RMSD, follows the TCRdock formulation [22,37]. MHC residues are matched between prediction and native by sequence alignment; a single ordinary-least-squares Kabsch superposition of the class I peptide-binding platform (or, jointly, of both class II peptide-binding domains) is applied to the entire predicted complex, and no further fit is performed. Cα RMSD is then computed over the eight CDR regions with CDR3 residues weighted 3 and all others weighted 1:

$$\mathrm{RMSD}_w = \sqrt{\frac{\sum_i w_i d_i^2}{\sum_i w_i}}$$

Because there is no second fit, this metric measures how accurately the TCR is docked on the pMHC.

The second, the framework-aligned CDR3 all-atom RMSD, measures the opposite quantity. For each chain independently, the predicted variable-domain framework (the IMGT-numbered variable domain excluding all four CDR regions) is superimposed on the native framework using matched non-hydrogen atoms, and all-heavy-atom RMSD is then computed over the corresponding CDR3 loop without further alignment, comparing only identically named atoms in sequence-matched, identity-matched residues. Cα-only versions are retained as quality-control columns.

2.6 Superposition-free CDR accuracy (lDDT)

Both metrics above answer their question through a superposition, and each superposition is a choice. The local distance difference test (lDDT) [38] removes that choice: a residue is scored by the fraction of its interatomic distances to the rest of the structure that the prediction reproduces, averaged over four tolerances (0.5, 1, 2 and 4 Å), counting only pairs closer than 15 Å in the reference and excluding pairs within one residue. It needs no alignment and no reference frame, and it is bounded in [0, 1].

lDDT was computed on Cα atoms, which is the quantity AlphaFold 2's pLDDT head was trained to predict [19] and requires no symmetric-side-chain correction; the all-atom view of the CDRs is already carried by the framework-aligned CDR3 RMSD. Residues were matched between prediction and native through the same manifest-sequence alignment, and the CDRs defined by the same ANARCI annotation, as the two RMSD metrics, so the residue sets behind the two families of number are identical. Every TCR region was scored twice, differing only in which atoms may act as contact partners: against every chain of the complex, which makes the score sensitive to where the TCR is docked, and against the CDR's own TCR chain alone, which removes the docking and leaves local loop geometry. Whole-chain, peptide, MHC and whole-complex lDDT are reported alongside for context.

2.7 Confidence and reference-free quality scores

Each method's own interface confidence (ipTM, introduced with AlphaFold-Multimer [39], and pTM where available) was taken from its output. Per-residue predicted lDDT (pLDDT), the residue-level confidence introduced with AlphaFold 2 [19], was read from the B-factor column of the same top-ranked prediction file that was scored, averaged over a residue's non-hydrogen atoms and then over a region; all four methods write it on the 0–100 scale here, and anything reported in [0, 1] is rescaled. Confidence is therefore compared with accuracy residue-for-residue over exactly the residue sets of Section 2.6, and the calibration error reported is the mean of pLDDT − lDDT over complexes. pDockQ [40] was computed per interface from the predicted structure and per-residue confidence, and averaged per complex. A published random-forest quality-tier classifier for TCR–pMHC models [26] was re-implemented against this benchmark's chain convention and scored on all 126 complexes; the subset of class I targets whose PDB identifiers do not appear in its training set (n = 27) is reported separately throughout.

2.8 Diffusion-step instrumentation and sampler timing

Two of the analyses below need to see inside the diffusion loop rather than only the structure it returns: the per-step denoising trajectories of Section 3.10, which compare how the two samplers approach their answer, and the sampler-economy comparison of Section 3.11, which asks how many diffusion steps and samples that answer actually needs. The benchmark's AlphaFold 3 predictions come from the AlphaFold Server (Section 2.3), which returns only final coordinates. Both experiments therefore required a second, local installation of AlphaFold 3, run on one RTX PRO 6000 with the MSAs and templates taken verbatim from the server bundles the benchmark's own predictions came from, so that the only difference from the scored predictions is the sampler instrumentation and the machine. ESMFold2 was likewise run from its local `biohub/ESMFold2` checkpoint rather than through the ESM SDK API. Per-frame accuracies from these runs are therefore not interchangeable with the benchmark's own DockQ values, and are used only to compare the two samplers with each other.

Both models' official samplers were patched observationally to retain the coordinate state after every production denoising step, changing no arithmetic: AlphaFold 3's `diffusion_head.sample` already computes and discards the per-step positions inside its scan, and the patch keeps them; ESMFold2's `DiffusionStructureHead.sample` holds its state in a local, so its installed source was rewritten at a unique anchor. Parity between instrumented and stock runs under the same seed was verified per model (mean $|\Delta\text{DockQ}| = 0.0008$, maximum 0.0122).

Sampler economy was measured on 30 complexes drawn with a frozen seed from the 126 scoreable set (28 class I, 2 class II, 397–416 residues), run serially on one idle RTX PRO 6000 with identical MSAs and templates taken verbatim from the AlphaFold Server bundles the benchmark's own predictions came from, and the same per-complex seed. Four tracks were run: 200 steps $\times$ 5 samples (default), 160 $\times$ 5, 200 $\times$ 1 and 160 $\times$ 1. All 30 complexes fall in AlphaFold 3's 512-token bucket, so JIT compilation and kernel autotuning happen once per track; both were charged to two warm-up predictions run and discarded before timing. Instrumentation costs ~2% and was run as a separate pass, so the baseline track is stock AlphaFold 3.

2.9 Statistics and software

Paired method comparisons use the Wilcoxon signed-rank test on the complexes both methods predicted. Correlations are Pearson and Spearman with bootstrap confidence intervals (10,000 resamples, fixed seed); regressions use ordinary least squares with HC3 robust standard errors.

3 Results

3.1 Composition of the benchmark set

The 126 scoreable complexes comprise 111 class I and 15 class II entries, 112 human and 14 mouse, released between 13 October 2021 and 29 July 2026. One hundred and twenty were determined by

X-ray crystallography and six by cryo-EM, with resolutions from 1.54 to 4.44 Å (median 2.56 Å, IQR 2.20–3.00 Å). They span 27 distinct MHC alleles — the most frequent being HLA-A*02 (29), H2-Db (10), HLA-A*11 (10), HLA-B*27 (9), HLA-A*24 (9) and HLA-E (7) — and 79 distinct peptide sequences of 8–15 residues (median 9).

3.2 Overall interface accuracy

Every method produced at least an acceptable model of every complex: Global DockQ never fell below 0.23 for any method on any of the 126 complexes (Figure 2A). Median Global DockQ was 0.751 for AlphaFold 3 (IQR 0.571–0.826), 0.736 for ESMFold2 (0.592–0.840), 0.708 for TCRmodel2 on its 111 class I targets (0.590–0.808) and 0.701 for Protenix (0.530–0.799).

The two leading methods are not distinguishable. Paired over all 126 complexes, AlphaFold 3 minus ESMFold2 gives a mean difference of −0.016 and a median difference of −0.002 (Wilcoxon $p = 0.68$). Both, however, separate from Protenix: AlphaFold 3 +0.039 ($p = 0.045$) and ESMFold2 +0.055 ($p = 0.0009$). TCRmodel2 was not separable from any of the three on its class I subset (all $p > 0.23$).

The CAPRI-class view (Figure 2B) tells the same story with a different emphasis. No method produced an incorrect model. ESMFold2 placed the largest fraction in the medium-or-better band (91.3%: 54.0% medium, 37.3% high), followed by TCRmodel2 (88.3%: 60.4% medium, 27.9% high), AlphaFold 3 (84.9%: 50.0% medium, 34.9% high) and Protenix (79.4%: 54.0% medium, 25.4% high). AlphaFold 3's slightly higher median therefore coexists with a slightly heavier acceptable-only tail than ESMFold2's — the two distributions cross.

This result runs counter to the current consensus [25,26]: a predictor that uses neither an MSA nor a structural template matches AlphaFold 3 on post-cutoff TCR–pMHC complexes.

3.3 Accuracy by interface

Global DockQ is weighted across all interfaces of the complex, several of which are modelled accurately by every method. Decomposing it by interface (Table 1) localises the differences between methods. The MHC–peptide interface is effectively solved by every method: median DockQ 0.940 (AlphaFold 3), 0.939 (ESMFold2), 0.933 (Protenix) and 0.931 (TCRmodel2), with the class II MHCα–peptide and peptide–MHCβ interfaces close behind (0.90–0.91). The TCRα–TCRβ pairing is also reliable (0.79–0.82). Everything that separates methods, and everything that separates easy complexes from hard ones, is in the four TCR-versus-pMHC interfaces, whose medians run 0.55–0.71. One class II interface is a partial exception: AlphaFold 3 models the MHC-IIα–MHC-IIβ heterodimer pairing itself at median DockQ 0.628, against 0.845 for ESMFold2 and 0.789 for Protenix, the single largest per-interface deficit for AlphaFold 3 in the table.

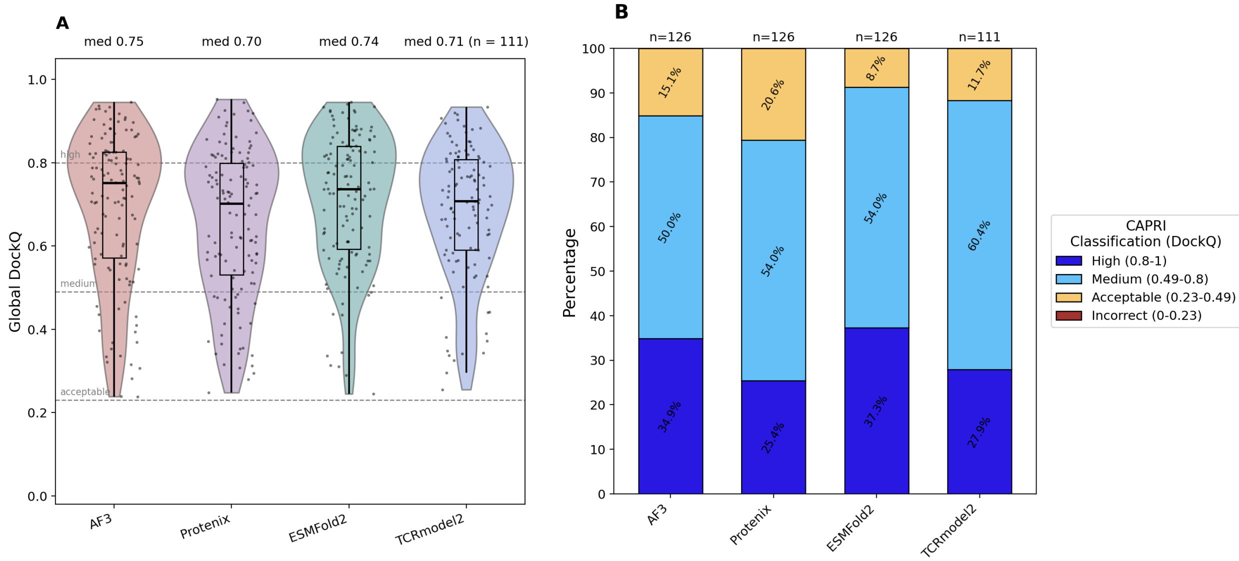

Figure 2: **Interface accuracy on 126 post-cutoff TCR–pMHC complexes.** (**A**) Distribution of Global DockQ per method; box, IQR and median; dots, individual complexes; dashed lines, CAPRI thresholds. (**B**) The same data as CAPRI classes, as a percentage of each method's own set. No method produced an incorrect model of any complex. TCRmodel2 covers the 111 class I complexes only, in both panels.

Table 1: **Median DockQ per interface.** Higher is better; the best method in each row is bold. n is the number of complexes that contribute the interface — an interface with no native contacts is not scored, which is why the peptide–TCR rows fall short of 126. TCRmodel2 predicts class I only (—). The final row collapses the four TCR-versus-pMHC interfaces of a complex to a single value before taking the median, and is the row the text refers to.

Interface	n	AlphaFold 3	Protenix	ESMFold2	TCRmodel2
Peptide in the groove					
MHC-I – peptide	111	**0.940**	0.933	0.939	0.931
MHC-IIα – peptide	15	0.904	**0.911**	0.905	—
peptide – MHC-IIβ	15	**0.909**	0.887	0.902	—
Chain pairing within a module					
TCRα – TCRβ	126	0.798	0.800	**0.821**	0.791
MHC-IIα – MHC-IIβ	15	0.628	0.789	**0.845**	—
TCR against pMHC					
MHC-I – TCRα	111	**0.672**	0.590	0.653	0.644
MHC-I – TCRβ	110	0.686	0.551	**0.696**	0.611
peptide – TCRα	120	**0.710**	0.605	0.676	0.641
peptide – TCRβ	122	0.687	0.633	**0.690**	0.681
MHC-IIα – TCRα	11	0.615	0.600	**0.714**	—
MHC-IIα – TCRβ	12	0.773	0.767	**0.839**	—
MHC-IIβ – TCRα	15	0.720	0.687	**0.819**	—
MHC-IIβ – TCRβ	15	0.719	0.519	**0.764**	—
All TCR–pMHC interfaces, per complex	126	0.692	0.621	**0.694**	0.632

Collapsing those four interfaces to one value per complex, median TCR-versus-pMHC DockQ is 0.694 for ESMFold2, 0.692 for AlphaFold 3, 0.632 for TCRmodel2 and 0.621 for Protenix. On this stricter reading the acceptable-or-better rate falls from 100% to 92.9% (ESMFold2), 90.1% (TCRmodel2), 87.3% (AlphaFold 3) and 84.1% (Protenix), and the high-accuracy rate is 28.6%, 27.8%, 19.8% and 17.5% respectively. Approximately one complex in ten is therefore still docked incorrectly by the best-performing methods.

3.4 TCR-specific accuracy: docking versus CDR3 loop conformation

The two TCR-specific metrics decompose that residual error (Table 2).

Table 2: **TCR-specific accuracy.** Cα or all-atom RMSD in ångströms, median (interquartile range); lower is better and the best method in each row is bold. Class I and II are pooled. The MHC-aligned metric applies one rigid superposition of the predicted peptide-binding platform onto the native and then measures all eight CDRs with CDR3 weighted 3, so it reports *docking*. The framework-aligned metrics fit each chain's variable-domain framework independently before measuring its CDR3 loop, so they report *loop conformation* with the docking removed.

Metric (å)	AlphaFold 3 $n = 126$	Protenix $n = 126$	ESMFold2 $n = 126$	TCRmodel2 $n = 111$
MHC-aligned, all CDRs, Cα	2.84 (1.61–5.56)	3.82 (2.10–7.47)	**2.75 (1.60–5.24)**	3.47 (1.99–5.16)
Framework-aligned CDR3α, all atom	1.94 (1.35–2.88)	2.33 (1.53–3.08)	**1.81 (1.25–2.59)**	2.02 (1.47–3.08)
Framework-aligned CDR3β, all atom	1.73 (1.17–2.49)	1.93 (1.40–2.60)	**1.56 (1.13–2.26)**	1.95 (1.42–2.70)

The MHC-aligned weighted all-CDR Cα RMSD, which measures docking with no second fit, has a median of 2.75 Å for ESMFold2, 2.84 Å for AlphaFold 3 and 3.82 Å for Protenix over the 126 complexes, and 3.47 Å for TCRmodel2 over its 111 class I predictions. The distributions are strongly right-skewed — means are 5.45 (ESMFold2), 6.34 (AlphaFold 3), 7.72 (Protenix) and 5.59 Å (TCRmodel2), and every method's maximum exceeds 50 Å — confirming that the aggregate is dominated by a minority of complexes in which the TCR is placed in an entirely wrong orientation, not by a uniform degradation.

Framework-aligned CDR3 RMSD, which removes docking error by construction, is far more uniform across methods. Median CDR3α all-heavy-atom RMSD is 1.81 Å (ESMFold2), 1.94 Å (AlphaFold 3), 2.02 Å (TCRmodel2) and 2.33 Å (Protenix); median CDR3β is 1.56 Å (ESMFold2), 1.73 (AlphaFold 3), 1.93 (Protenix) and 1.95 (TCRmodel2). The spread across methods for loop conformation (0.5 Å) is an order of magnitude smaller than the spread across complexes for docking (tens of ångströms), and the maxima are bounded — no method exceeds 7.9 Å on CDR3α or CDR3β. In other words, all four methods build approximately the same quality of CDR3 loop, and differ almost entirely in where they put it. This is consistent with Shi et al.'s observation that CDR3 loops and docking orientation are separate, and separately unsolved, problems [27].

Two secondary observations follow from Table 2. First, ESMFold2's advantage is concentrated in the CDR3β loop and in class II: its class II medians are 1.76 Å (all-CDR), 1.82 Å (CDR3α) and 1.34 Å (CDR3β), against AlphaFold 3's 2.37, 2.37 and 2.05 Å. Second, restricting a model's template search to TCR and MHC structures does not spare it the catastrophic failures. On the 111 class I complexes TCRmodel2's all-CDR distribution is less dispersed than AlphaFold 3's or Protenix's (SD 7.59 Å against 9.31 and 9.34) but not than ESMFold2's (7.03 Å), and its maximum is the same

52.7 Å as AlphaFold 3's. Its advantage over the general models is a lower mean (5.59 Å against AlphaFold 3's 6.18 on the same complexes), not an absence of wrong orientations.

3.5 Superposition-free CDR accuracy and predicted local confidence

Both metrics of Section 3.4 answer their question through a superposition, and each superposition is a choice. lDDT [38] removes that choice: it scores a residue by how many of its interatomic distances to the rest of the structure the prediction reproduces, so it needs no alignment and no reference frame. It is also the quantity every one of these methods predicts per residue and reports as pLDDT, which makes it the one accuracy measure directly comparable with the models' own confidence. We computed Cα lDDT over the same eight CDR regions, on the same top-ranked predictions, with two contact sets: the whole complex, and the CDR's own TCR chain alone (Table 3, Figure 3).

Table 3: **CDR lDDT and the models' own pLDDT.** Median (interquartile range), class I and II pooled. lDDT is superposition-free and bounded in [0, 1]; higher is better and the best method in each accuracy row is bold. The contact set is what the score is allowed to see: *complex* admits every chain, so the score falls when the TCR is docked in the wrong place, while *own chain* admits only the TCR chain the loop sits on, which removes the docking and leaves local geometry. The last row is not an accuracy: it is each method's own per-residue confidence over the same residues, rescaled to [0, 1], and is bold nowhere because a higher value is only better if it is true.

Region	Contact set	AlphaFold 3 $n = 126$	Protenix $n = 126$	ESMFold2 $n = 126$	TCRmodel2 $n = 111$
All CDRs	complex	0.866 (0.79–0.91)	0.844 (0.78–0.89)	**0.872 (0.82–0.91)**	0.852 (0.81–0.89)
All CDRs	own chain	0.921 (0.88–0.95)	0.916 (0.88–0.94)	**0.932 (0.89–0.96)**	0.920 (0.89–0.94)
CDR3α	complex	0.824 (0.71–0.90)	0.787 (0.72–0.87)	**0.835 (0.73–0.89)**	0.792 (0.73–0.88)
CDR3β	complex	0.842 (0.77–0.92)	0.824 (0.74–0.90)	**0.849 (0.78–0.93)**	0.819 (0.77–0.89)
Framework	complex	**0.961 (0.94–0.97)**	0.958 (0.95–0.97)	0.960 (0.95–0.97)	**0.961 (0.95–0.97)**
Whole complex	complex	0.935 (0.91–0.96)	0.931 (0.91–0.95)	0.940 (0.92–0.95)	**0.942 (0.92–0.95)**
Confidence, not accuracy					
Mean CDR3 pLDDT	—	0.874 (0.80–0.92)	0.890 (0.84–0.94)	0.784 (0.72–0.82)	0.826 (0.78–0.86)

Within a method, all-CDR lDDT in the whole-complex frame is very nearly a restatement of docking quality: it correlates with Global DockQ at Pearson r = 0.934 (AlphaFold 3), 0.915 (ESMFold2), 0.907 (Protenix) and 0.904 (TCRmodel2), and with the MHC-aligned weighted all-CDR RMSD of Section 3.4 at Spearman $\rho = -0.911$, -0.897, -0.887 and -0.827. Between methods it says much less. The four medians span 0.844 to 0.872 — a range of 0.028 — for CDRs whose MHC-aligned RMSD medians span 2.75 to 3.82 Å. lDDT is local and bounded, and a TCR placed in an entirely wrong orientation keeps most of the interatomic distances within its own chains, so the catastrophic minority that dominates the RMSD means is compressed into the lower whisker of Figure 3A rather than allowed to run to 50 Å. lDDT therefore ranks complexes well and separates methods poorly, and the two should not be substituted for one another.

Restricting the contact set to the CDR's own TCR chain reproduces Section 3.4's conclusion without any superposition at all. All-CDR lDDT rises to 0.916–0.932, a spread of 0.016 across four methods, and its correlation with Global DockQ falls from about 0.91 to 0.680 (AlphaFold 3), 0.674 (ESMFold2), 0.600 (TCRmodel2) and 0.557 (Protenix). The loops are built to nearly the same local standard everywhere; what differs is where they are put.

pLDDT ranks CDR3 accuracy better than any of the global confidence scores rank whole-complex accuracy, reproducing on four methods and a superposition-free measure what Lu et al. reported for AlphaFold 3 against CDR3 RMSD [25]. Pooling CDR3α and CDR3β, Pearson r between mean CDR3 pLDDT and CDR3 lDDT is 0.837 for ESMFold2, 0.750 for Protenix, 0.728 for TCRmodel2 and 0.724 for AlphaFold 3 — the same ordering, and for ESMFold2 a higher value, than the ipTM correlations of Section 3.7. Its *calibration*, however, changes sign across the variable domain (Figure 3C). AlphaFold 3, Protenix and TCRmodel2 understate the framework (mean pLDDT − lDDT = −0.027, −0.007 and +0.004) and overstate CDR3 (+0.035, +0.084 and +0.011); Protenix is the most overconfident loop-builder of the four, and the only method whose CDR3 pLDDT exceeds even its own-chain CDR3 lDDT. ESMFold2 is uniformly conservative — −0.100 on the framework rising to −0.058 on CDR3 — which is the per-residue counterpart of the ipTM intercept in Section 3.7: its confidence is on a lower scale but better ordered. Measured against the own-chain lDDT instead, every method is underconfident on CDR3 (−0.035, +0.005, −0.125 and −0.060). Predicted lDDT therefore sits between the loop's local quality and the quality it achieves once docked: the confidence head anticipates part of the docking error, but not the part that matters.

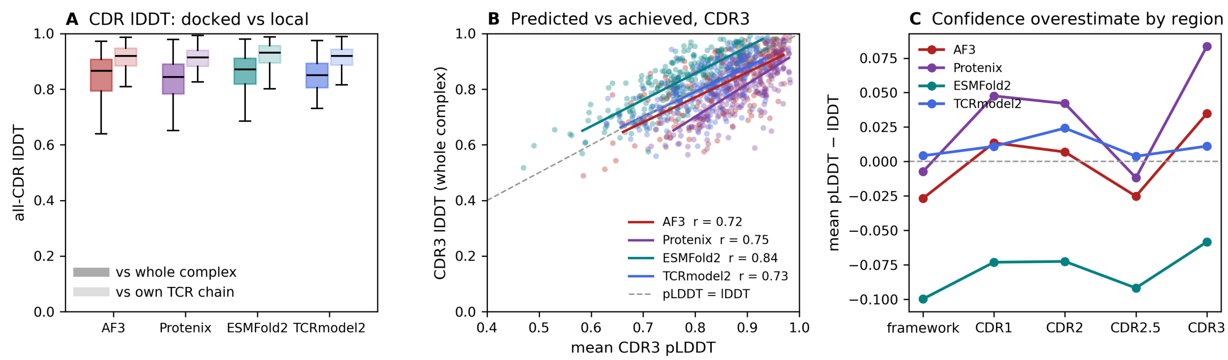

Figure 3: **Superposition-free CDR accuracy and the models' own confidence.** (**A**) All-CDR Cα lDDT with the whole complex as contact partners (dark) and with the CDR's own TCR chain only (pale). Removing the docking raises every method by a similar amount and compresses the differences between them. (**B**) Mean CDR3 pLDDT against the CDR3 lDDT actually achieved, CDR3α and CDR3β pooled, with per-method linear fits and Pearson r; the dashed line is perfect calibration. (**C**) Mean pLDDT − lDDT by region. Three of the four methods cross from understating the framework to overstating CDR3; ESMFold2 is conservative everywhere.

3.6 Effect of MHC class

Class II is not harder than class I on this benchmark (Figure 4B). Median Global DockQ for class II versus class I is 0.752 versus 0.750 (AlphaFold 3), 0.742 versus 0.736 (ESMFold2) and 0.713 versus 0.698 (Protenix); every difference is well inside the spread, and the class II subset is small (n = 15). ESMFold2's class II mean (0.743) is its highest subgroup mean. We note this mainly as a caution: with 15 class II complexes, this benchmark cannot resolve a class effect of the size other studies report, and the TCRmodel2 comparison is class I only by construction.

3.7 Confidence calibration and reference-free quality scores

Interface confidence predicts accuracy for all three general methods, but ESMFold2's ipTM does so substantially better (Figure 4A). Pearson r between ipTM and Global DockQ is 0.793 for ESMFold2 (Spearman $\rho = 0.777$), against 0.660 for AlphaFold 3 ($\rho = 0.654$) and 0.657 for Protenix ($\rho = 0.679$). The regression lines also differ in intercept: ESMFold2 reaches a given DockQ at a systematically lower ipTM than the other two, so its confidence is both more conservative and better ordered. pDockQ shows the same ranking with lower correlations overall (ESMFold2 r = 0.676, Protenix 0.611, AlphaFold 3 0.577).

A learned quality model does not improve on this. The published random-forest quality-tier classifier for TCR–pMHC models [26], re-implemented against this benchmark's chain convention and scored on the same 126 complexes, reaches r = 0.662 ($\rho = 0.674$) against Global DockQ — indistinguishable from AlphaFold 3's own ipTM at 0.660 (0.654), the score it was built to improve on. On the 27 class I targets whose PDB identifiers are absent from its training set the two remain indistinguishable. The reference-free signal that does beat both is not a quality model at all (Section 3.9).

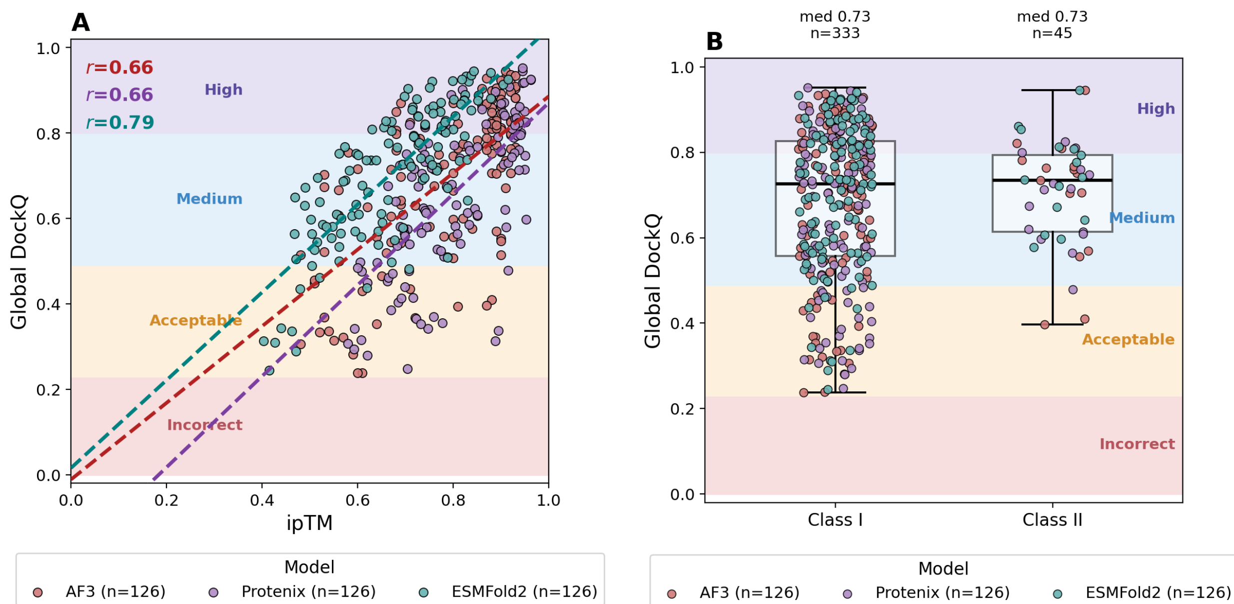

Figure 4: **Determinants of interface accuracy.** (**A**) Global DockQ against each method's own ipTM, with per-method linear fits and Pearson r. ESMFold2's confidence is both more conservative — it reaches a given DockQ at a lower ipTM — and better ordered. (**B**) Global DockQ by MHC class. Shaded bands are the CAPRI quality classes.

3.8 What each method takes from the pre-cutoff record

Every target was audited against the 259 TCR–pMHC complexes released on or before 30 September 2021, using the reference set of Section 2.1 and four criteria: release date; TCR redundancy (concatenated Vα+Vβ identity $\geq$ 90%, or either chain $\geq$ 95%, by global alignment with $\geq$ 80% coverage); all-chain similarity to a single pre-cutoff complex; and whole-complex structural similarity by Foldseek-Multimer. No target breaches the cutoff — the earliest release is 13 October 2021, though 36 targets were *deposited* before it. Sequence redundancy is another matter: 42 targets carry a redundant TCR, 43 have every corresponding chain above 40% identity to one pre-cutoff complex, and 10 of those clear 70% on every chain, two of them (7RTR versus 7N1F, 9NDS versus 4FTV) being the same complex solved twice. Within the benchmark the 126 targets collapse to 66 independent clusters.

Accuracy rises with that similarity, but through a narrow channel (Figure 5A–C). Peptide and MHC identity to the closest pre-cutoff complex predict almost nothing (Spearman $\rho = +0.04$ to +0.20); TCR Vβ identity predicts a little for every method (+0.29 to +0.37); and the concatenated TCR identity separates the methods — +0.12 for AlphaFold 3 and +0.10 for Protenix against +0.31 for ESMFold2 and +0.22 for TCRmodel2. The strongest correlate of all is not sequence but structure: the multimer TM-score between the target's own experimental structure and the nearest pre-cutoff complex reaches $\rho = +0.38$ (AlphaFold 3), +0.50 (Protenix), +0.45 (ESMFold2) and +0.55 (TCRmodel2).

Reference quality acts on the same numbers from the other side (Figure 6D). Accuracy declines with worsening reference resolution for every method, but not equally: Pearson r between resolution and Global DockQ is −0.292 for AlphaFold 3 ($p = 0.001$), −0.266 for ESMFold2 ($p = 0.003$), −0.238 for TCRmodel2 ($p = 0.012$) and −0.184 for Protenix ($p = 0.040$, Spearman $p = 0.065$, not significant). Part of this is unavoidable — a lower-resolution reference is a noisier target — but the ordering is informative: the two methods most sensitive to reference quality are the two most accurate ones, which is what one expects if the residual error in the best models is approaching the coordinate uncertainty of the references themselves.

Sorting the targets by audit class makes the size of the effect concrete (Figure 5E–F). Every method is worst on the 42 targets clean of every criterion and best on the 10 near-duplicates, monotonically across the four classes. The median Global DockQ gained on the near-duplicates relative to the clean set is +0.16 for AlphaFold 3 (95% bootstrap CI +0.00 to +0.32), +0.26 for Protenix (+0.05 to +0.41), +0.23 for ESMFold2 and +0.16 for TCRmodel2. Reported accuracy on this benchmark, and on any benchmark of comparable construction, is raised by its redundant tail.

Two observations keep this from being read as a memorisation result. ESMFold2 shows the effect as strongly as any method — the largest concatenated-TCR correlation and a +0.23 near-duplicate gain — while receiving no MSA and no structural template, only the query sequence, so whatever produces the gradient is not template retrieval. And the strongest correlate, structural resemblance to the pre-cutoff record, is confounded with how canonical a complex is: canonical TCR–pMHC geometries are both densely represented in the old record and intrinsically easier to predict, so that correlation cannot separate memorisation from typicality. The measures that could separate them — peptide and MHC identity — are the ones near zero.

What the redundancy does change is how method comparisons should be read. Filtering at $\leq$ 95% maximum TCR identity retains 69 targets and barely moves absolute accuracy (AlphaFold 3 mean DockQ 0.585 $\rightarrow$ 0.547; Protenix 0.515 $\rightarrow$ 0.506), but it dissolves the AlphaFold 3–over–Protenix result: the paired advantage falls from +0.063 ($p = 0.031$) to +0.027 ($p = 0.40$), and at $\leq$ 90% it reverses sign. Below 95% the retained samples (14, 7 and 1 targets) are too small to conclude anything either way. The AlphaFold 3 advantage over Protenix on this benchmark is therefore carried by targets with a near-duplicate TCR in the pre-cutoff PDB. It is also why the primary benchmark is not built on a TCR-identity filter: only 7 of 126 targets fall below 70% identity, and those 7 are not hard targets but mostly unusual or non-human TCRs. Applying every audit criterion at once would leave 42 targets, 25 after collapsing the within-benchmark clusters — too few, and too depleted of class II (4 complexes), to carry the comparisons this paper makes.

Excluding only the 10 near-duplicates and the 42 TCR-redundant targets leaves 84, which we report as the redundancy-controlled subset rather than the primary set.

A final caution against reading availability as an answer key: 7RTR has a 100%-identical complete pre-cutoff complex in 7N1F, and AlphaFold 3 still scores it at 0.36, CAPRI acceptable. The negative MHC coefficient in a pooled regression over these covariates is likewise an artefact of no variance rather than evidence that MHC familiarity hurts — MHC identity has median 100% and IQR 99.4–100%, because 111 of 126 targets are class I HLA alleles whose peptide-binding domain is in the PDB many times over.

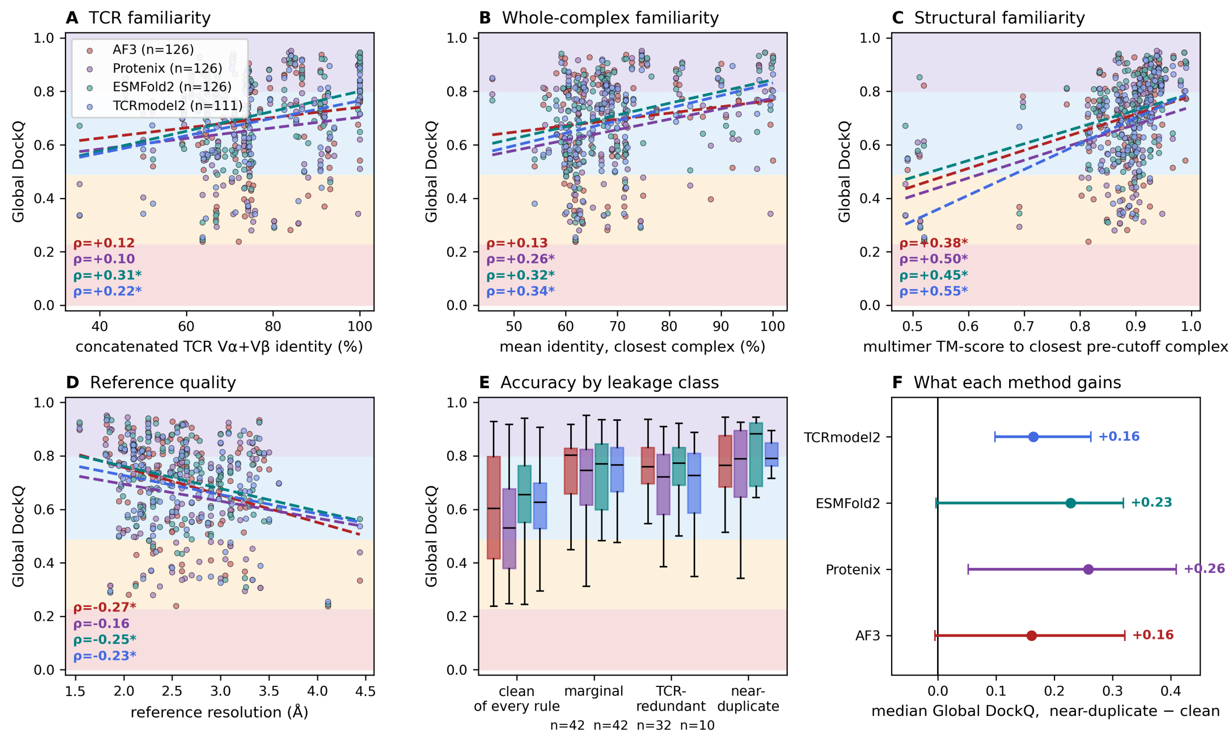

Figure 5: **What accuracy tracks besides the model.** (**A**) Global DockQ against concatenated TCR Vα+Vβ identity to the closest pre-cutoff complex, (**B**) against mean identity across all chains of the closest complex, (**C**) against the multimer TM-score between the target's own experimental structure and the closest pre-cutoff complex, and (**D**) against the resolution of the experimental reference; dashed lines are per-method linear fits and ρ is Spearman, starred where $p < 0.05$. (**E**) Accuracy by audit class. (**F**) Median Global DockQ gained on the 10 near-duplicates relative to the 42 targets clean of every criterion, with 95% bootstrap intervals. Shaded bands in A–E are the CAPRI quality classes.

3.9 Structural divergence between AlphaFold 3 and ESMFold2

Equal accuracy is not the same as equal predictions. Across the 126 complexes the median Cα RMSD between the AlphaFold 3 and ESMFold2 predictions of the same FASTA is 1.35 Å, but the distribution runs to 13.9 Å, and the disagreement is anatomically specific. The two models agree closely on everything except where the TCR sits: median peptide RMSD in the MHC frame is 0.44 Å and median peptide conformation RMSD is 0.33 Å, and the median internal RMSD of the TCR module itself is 0.74 Å — but the median MHC–TCRα and MHC–TCRβ interface RMSDs are 1.62 and 1.63 Å, and the rigid-body decomposition gives a median TCR rotation of 11.1° with a tail reaching 177°. On 20 of 126 complexes the two models place the TCR more than 30° apart, and on 10 of them more than 90° apart, while building the same pMHC and the same TCR fold. The disagreement is a docking disagreement, not a folding one.

The disagreement is also bimodal rather than graded. Of the 126 complexes, 106 place the TCR within 30° of each other and only 12 exceed 60°; there is essentially nothing in between, and those 12 are exactly the 12 largest cross-model RMSDs. Superposing each prediction's MHC on the native's and measuring its TCR against the native's separates the two models on these cases: among the 20 complexes that disagree by more than 30°, AlphaFold 3 is the misplaced one on 12, ESMFold2 on 4, and both are misdocked on 4. Over the 10 largest disagreements the median Global DockQ is 0.335 for AlphaFold 3 and 0.581 for ESMFold2, and only 2 of the 10 have both models at medium-or-better accuracy, against 101 of the remaining 116. Where the two architectures disagree about docking, at least one of them has failed, and it is usually AlphaFold 3.

Two geometrically distinct failures produce the disagreement. In the first, the misplaced TCR is rotated 100–177° *and* translated 50–63 Å: it is not docked on the groove at all but on another surface of the MHC, still burying 55–150 Cα contacts there while making no contact with the peptide (9EJG, 8RYO, 7RK7, 7PB2; the fifth such case, 9EJH, is the inverse — its native TCR is the one docked away from the peptide, and AlphaFold 3 builds the canonical peptide-centred pose instead). In the second, the TCR still straddles the groove and still contacts the peptide, but is turned around on it: 70–177° of rotation with the centre of mass only 4–12 Å from where the other model puts it, a reversal of docking polarity rather than a relocation (9RU5, 8ES9, 7RM4, 7RRG, 9NW2).

The relocations have a specific and avoidable cause. Projecting the TCR centre on the axis from the MHC centre to the peptide centre places the native TCR 28.8 Å above the peptide-binding platform in the median complex, and no class I native below 14 Å. Three predictions — 8RYO, 7RK7 and 7PB2, all AlphaFold 3, all class I — put the TCR 24–26 Å *below* the platform, docked on the β-sheet floor that in the intact molecule is covered by the α3 domain and β2-microglobulin and that the domain truncation of Section 2.2 leaves exposed. On these three complexes AlphaFold 3 scores 0.28, 0.24 and 0.31 while ESMFold2, given the same truncated input, scores 0.91, 0.53 and 0.63 and places the TCR within 3–19° of the native. Part of AlphaFold 3's tail of docking failures is therefore an artefact of presenting it with a non-physiological surface, not a failure to recognise the correct one — a caveat that applies to any truncated-input benchmark, and one ESMFold2 appears not to share.

That disagreement is a strong, entirely reference-free predictor of accuracy (Figure 6A). Cross-model Cα RMSD correlates with AlphaFold 3's Global DockQ at Spearman $\rho = -0.756$ and with ESMFold2's at $\rho = -0.670$, and with the *worse* of the two at $\rho = -0.884$ ($p = 1.1 \times 10^{-42}$). For comparison, on the same 126 complexes AlphaFold 3's own ipTM predicts that worse-of-two value at $\rho = +0.604$ and ESMFold2's at $\rho = +0.639$. Cross-architecture agreement is therefore a better native-free quality signal than either model's internal confidence, than pDockQ, and than the published random-forest quality-tier classifier — and it requires no confidence head, no training, and no calibration.

By divergence quartile (Figure 6B), median Global DockQ falls from 0.860 (AlphaFold 3) and 0.862 (ESMFold2) in the most-agreeing quartile to 0.518 and 0.564 in the most-divergent one. Notably, the two models mostly fail on the *same* complexes: both quartile trends are parallel, and among the 20 most divergent complexes, 4 are failed by both at the medium threshold while 12 are solved by exactly one.

The residual complementarity is nonetheless substantial (Figure 6C). At DockQ ≥ 0.49, AlphaFold 3 succeeds on 107 of 126 complexes and ESMFold2 on 115; the two together succeed on 119 (103 both, 4 AlphaFold 3 only, 12 ESMFold2 only, 7 neither). Pooling an MSA-based and a language-model-based predictor therefore raises the medium-or-better rate from 85% or 91% to 94% — echoing the pooling gains Yin et al. report for antibody–peptide complexes [28] — and, because disagreement itself flags the failures, the pool comes with a built-in triage signal.

Two related class II entries illustrate the pattern concretely. 7SG1 and 7SG2 are HLA-DQ2 complexes with closely related gluten-derived peptides. On 7SG2 both models are correct (AlphaFold 3 0.706, ESMFold2 0.642) and their predictions differ by 2.72 Å with a 14° TCR rotation. On 7SG1 the models diverge by 3.56 Å with a 26° rotation, and AlphaFold 3 drops to 0.396 while ESMFold2 holds at 0.736. Both models report high confidence on both complexes (AlphaFold 3 ipTM 0.87 and 0.92; ESMFold2 0.81 and 0.83) — the confidence heads do not see the failure, and the disagreement does.

Rendering three complexes side by side makes the two failure modes and the successful case directly visible (Figure 7). Colour identifies the *model*, not the chain: the experimental structure is gray, AlphaFold 3 firebrick red and ESMFold2 teal, so the MHC, peptide, TCRα and TCRβ of one model all carry that model's colour. The first three columns show each complete complex, chains A–D. In the fourth column the two predicted MHC chains are hidden and only the experimental MHC chain A is kept, as the common platform on which all three structures were superposed; the three TCRα and TCRβ pairs and the three peptides (sticks) are all retained, with the experimental structure drawn partially transparent. Because the superposition is on the MHC platform, any separation between the TCRs in that column is a difference in predicted placement on the same pMHC, not a difference in viewpoint.

8RYO is the global docking failure. In the experimental structure the gray TCR$\alpha\beta$ module sits above the peptide-binding groove, and ESMFold2 reproduces that arrangement — the teal TCR occupies the same region as the gray one, 2.6° and 0.1 Å from it. AlphaFold 3 instead places the red TCR on the opposite face of the platform, below the groove against the β-sheet floor, 162° rotated and 53 Å away. The individual domains remain protein-like; only their quaternary arrangement is wrong. In the fourth column the retained gray MHC separates two entirely different TCR positions, and the peptide sticks mark where the true groove is. The conclusion is not a peptide-conformation difference (0.30 Å) but a wrong docking surface, and Global DockQ follows: 0.281 against 0.907.

9GV7 is the quieter and more cautionary case. Both models place the TCR in a similar orientation — 44° apart from each other — but that shared solution is displaced from the experimental one, by 160° for AlphaFold 3 and 116° for ESMFold2. In the fourth column the red and teal TCRs overlap each other more closely than either overlaps the gray, their footprints are shifted across the peptide–MHC surface, and AlphaFold 3's TCRα is displaced far enough that part of it leaves the groove region altogether. Neither model is right (0.315 and 0.289), and their agreement is not evidence that either is. 9GV7 also carries an 11-residue peptide, SLAGGLDDMKA, with a central bulge, and it is the one complex of the three where the predicted peptide conformations neither coincide with the experimental one nor with each other. The two models disagree about the peptide by 1.25 Å, against 0.30 Å in 8RYO and 0.16 Å in 43RH, and both deviate from the native across the bulge itself (Cα RMSD over P4–P8: 1.85 Å for AlphaFold 3, 2.57 Å for ESMFold2, against 0.2–0.6 Å at either terminus, which stays anchored). The two differences are related: a peptide the models build differently is a peptide whose surface presents differently to the TCR.

43RH is the positive control, and the hardest complex in the benchmark that both models solved. It is a non-classical mouse H2-Q9 complex whose TCR is only 64% identical to any pre-cutoff sequence, whose peptide has no pre-cutoff hit at all, and for which no pre-cutoff complex is similar enough to serve as a template — and its native is a 3.33 Å structure. Both models nonetheless land within 5° and 10° of the experimental TCR placement and 0.55 Å of each other (Global DockQ 0.847 and 0.767). In the fourth column all three TCR$\alpha\beta$ modules and all three peptides superimpose, which is what agreement looks like when it is also correct — the control that makes the 9GV7 panel interpretable.

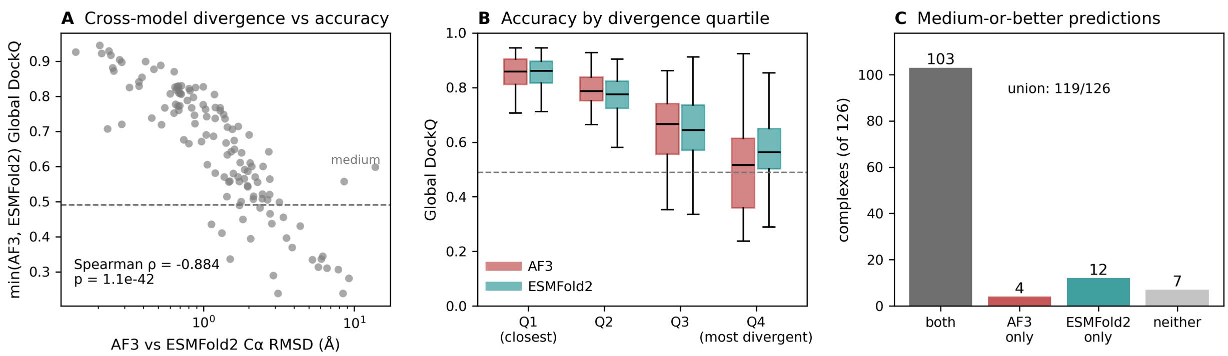

Figure 6: **Structural divergence between the AlphaFold 3 and ESMFold2 predictions.** (**A**) Cα RMSD between the AlphaFold 3 and ESMFold2 predictions of the same sequence against the worse of their two Global DockQ scores. (**B**) Global DockQ by quartile of that divergence; the two methods degrade together. (**C**) Complexes reaching medium-or-better accuracy: individually, jointly, and neither.

3.10 Denoising trajectories of the two samplers

Instrumenting both diffusion samplers on the same complexes shows that the equality of outcome conceals very different dynamics (Figure 8A). AlphaFold 3 begins denoising from an initial noise scale of ~2,560 Å and is still ~44 Å from the native at the midpoint of its 200-step schedule; ESMFold2 begins at ~256 Å and is already within ~5.7 Å at its own midpoint. Measured as the fraction of the schedule at which the trajectory comes permanently within 1 Å of its own final frame, AlphaFold 3 converges at 73% of its steps and ESMFold2 at 58%. AlphaFold 3 spends most of its schedule in a regime where, at the scale the noise schedule sets, no structure has settled at all; ESMFold2 commits early and refines.

The same trajectories say when each sampler actually builds the interface, which is the part of the structure this benchmark scores (Figure 8B). Counting the fraction of native interface contacts present at each step, the median AlphaFold 3 run has no native contact at all until 58% of its schedule and acquires most of its interface between 60% and 80%; the median ESMFold2 run has its first native contact at 40% and is within a few percent of its final interface by 60%. The two samplers therefore differ not only in when coordinates stop moving but in when the interface exists to be scored: ESMFold2 spends the back half of its schedule refining an interface it has already made, and AlphaFold 3 spends the front half without one.

The same behaviour at scale — the 30 complexes × 5 samples of the sampler-economy runs, 30,000 recorded frames — puts AlphaFold 3's convergence in a narrow window. Mean Cα RMSD to the

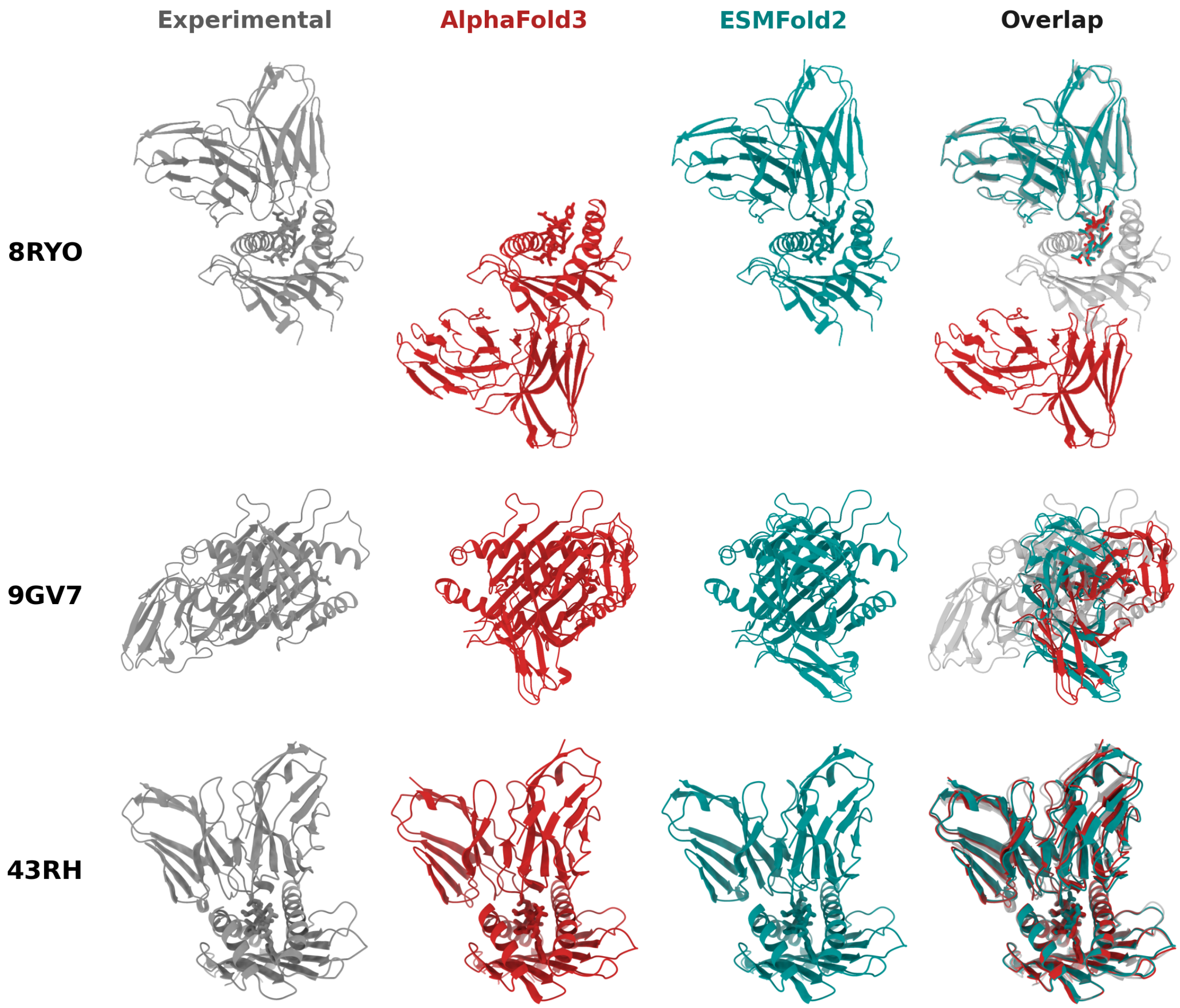

Figure 7: **Experimental, AlphaFold 3 and ESMFold2 TCR–pMHC structures for three complexes.** Experimental structures are gray, AlphaFold 3 predictions firebrick red and ESMFold2 predictions teal; colour denotes the model, not the chain. Chains A, B, C and D are the MHC-I $\alpha1$–$\alpha2$ platform, the peptide, the TCRα variable domain and the TCRβ variable domain. The first three columns show each complete complex. In the *Overlap* column the predicted MHC chains are hidden and the experimental MHC is retained as the common reference on which all three structures were superposed, with every TCRα, TCRβ and peptide (sticks) still shown and the experimental structure partially transparent; TCR separation in that column is therefore a difference in predicted placement, not in viewpoint. In 8RYO ESMFold2 approximately reproduces the experimental TCR placement while AlphaFold 3 docks the TCR on the opposite face of the MHC platform. In 9GV7 the two models predict similar TCR placements but both are displaced from the experimental interface, and the conformation of the bulged 11-residue peptide also differs. In 43RH — a template-free, non-classical H2-Q9 complex — both models reproduce the experimental placement and all three structures superimpose.

final frame of the same sample falls below 2.0 Å at step 150 and below 1.0 Å at step 158; every complex is within 1.0 Å by step 159; accuracy against the native plateaus at 3.37 Å around step 163; and mean RMSD to the final frame falls below 0.5 Å at step 164. Four independent criteria land between steps 150 and 164 of 200. For roughly the first 140 steps the state is noise at the scale the schedule sets — AlphaFold 3's schedule holds σ above 5 Å until ~70% of the way through.

3.11 Effect of AlphaFold 3 sampler settings on accuracy and runtime

If the trajectory has effectively converged by step ~160, the last 40 steps and the four extra diffusion samples should be nearly free to remove. They are (Figure 8C, Table 4). Over the same 30 complexes, same MSAs, same seeds, run serially on one idle GPU:

Table 4: **AlphaFold 3 sampler economy on 30 complexes.** Same complexes, same MSAs and templates, same per-complex seed, run serially on one idle RTX PRO 6000. GPU seconds are mean ± SD over the 30 complexes and exclude a constant 3.3 s of CPU featurisation. Δ is the paired change in mean Global DockQ against the baseline track, with a 95% paired-bootstrap interval; p is a paired test against the same baseline. Every interval includes zero.

Track	Steps × samples	GPU s	Speed-up	Global DockQ	Δ (95% CI)	p
Baseline	200 × 5	10.04 ± 0.03	1.00×	0.7212	—	—
Fewer steps	160 × 5	8.79 ± 0.03	1.14×	0.7143	−0.007 (−0.026, +0.007)	0.43
One sample	200 × 1	6.02 ± 0.01	1.67×	0.7143	−0.007 (−0.021, +0.006)	0.34
Both	160 × 1	5.57 ± 0.01	**1.80**×	0.7109	−0.010 (−0.032, +0.007)	0.33

Every confidence interval straddles zero, no track loses a complex from the acceptable band (30/30 everywhere), and the high-accuracy fraction moves from 0.367 to 0.333 — one complex. Adding the constant 3.3 s of CPU featurisation gives the end-to-end figure: 13.4 s → 8.9 s per complex, or 401 s → 267 s for all 30. Note that the 160-step track is not the 200-step trajectory stopped early — AlphaFold 3 re-discretises the same σ range into 160 coarser steps — so this is an experiment testing the convergence analysis, not a restatement of it.

The speed-up saturates at 1.8× for a structural reason. Fitting t = trunk + c × (steps × samples) across all 120 timed predictions gives a fixed trunk of 4.86 s and 5.08 ms per sample-step (R^2 = 0.995). Half of a default AlphaFold 3 run on a complex of this size is the Pairformer trunk and its 10 recycles, which neither shortcut touches; driving diffusion cost to zero would give at most 1.9×. Dropping from five samples to one also costs the ranking step — AlphaFold 3 normally returns the best of five by ranking score — and on this sample that selection is worth −0.007 DockQ, which is not detectable at n = 30. Woods et al. reached the same conclusion from the other direction, running AlphaFold 3 on TCR triads with five seeds and a single diffusion sample because extra seeds bought more accuracy than extra samples [37].

4 Discussion

4.1 Multiple sequence alignments are not required for TCR–pMHC modelling

The clearest result reported here is a negative one: on 126 post-cutoff complexes, a diffusion predictor conditioned only on a protein language model matched AlphaFold 3 running with server-

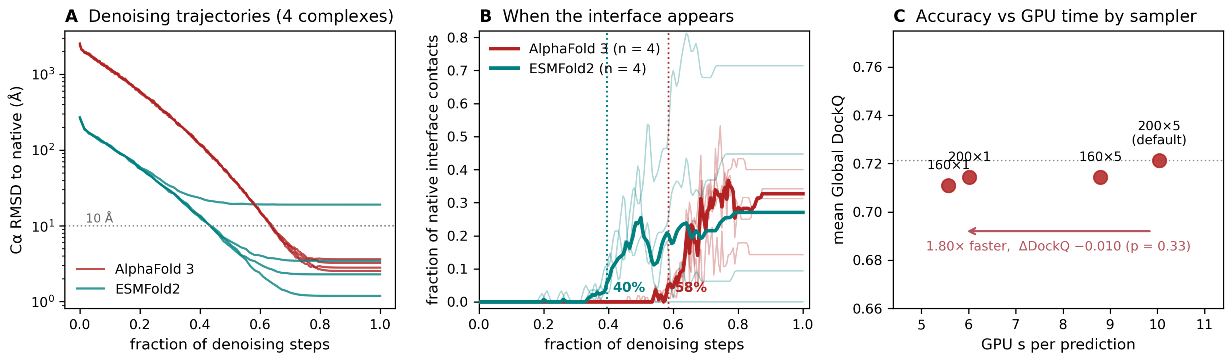

Figure 8: **Denoising trajectories and AlphaFold 3 sampler economy.** (**A**) Cα RMSD to the native along the denoising trajectory for four complexes traced in both models; AlphaFold 3 (200 steps, initial $\sigma \approx 2{,}560$ Å) is still tens of ångströms out at its midpoint, ESMFold2 (68 recorded steps, initial $\sigma \approx 256$ Å) is within a few ångströms. (**B**) Fraction of the native interface contacts present along the same trajectories; thin lines are individual complexes, thick lines the median of four, and the dotted verticals mark where the median run acquires its first native contact. (**C**) GPU time against mean Global DockQ for four sampler configurations on the same 30 complexes.

supplied MSAs and templates (median Global DockQ 0.736 vs 0.751, paired p = 0.68), and exceeded it in the fraction of medium-or-better models (91.3% vs 84.9%). This differs from the current consensus [25,26]. The discrepancy appears to be generational rather than contradictory: the PLM predictors in those comparisons are ESMFold-generation models whose structure module descends from AlphaFold 2, whereas ESMFold2 pairs a 6-billion-parameter language model with a diffusion structure head. The most informative data point in Lu et al. is that their best-performing PLM method was the one built on an AlphaFold-3-style architecture [25]; the present result is the same observation with the architecture gap closed further.

Four further observations in this benchmark point the same way, and none of them is a DockQ median. ESMFold2's confidence is the best calibrated of the four methods (ipTM against Global DockQ, Pearson r = 0.793 versus 0.660 for AlphaFold 3), so the MSA-free model is not merely as accurate but better at knowing when it is right. Its strongest subgroup is class II (mean Global DockQ 0.743), the setting in which paired alignments are hardest to build. It is the only method that does not mistake the exposed β-sheet floor left by domain truncation for a docking surface, scoring 0.91, 0.53 and 0.63 on the three complexes where AlphaFold 3 scores 0.28, 0.24 and 0.31. And it returns all of this in 5.2 s per complex against 13.4 s, with no alignment database, no homology search and no template lookup anywhere in the loop — the stage that dominates wall-clock time at repertoire scale, and the one that cannot be run at all for a synthetic or heavily engineered receptor with no natural homologues. On this benchmark the absence of an MSA is not a handicap accepted in exchange for speed: it costs nothing measurable and removes the pipeline's least tractable dependency.

Why this should hold for this problem in particular is worth stating explicitly: TCR variable domains and MHC platforms are among the most heavily sequenced and most heavily crystallised folds in existence, so an MSA adds little that a large language model has not already internalised, whereas the quantity an MSA cannot supply — the relative orientation of two domains with no coevolutionary history with one another — is precisely what remains unsolved.

4.2 Docking, not loop conformation, is the unsolved problem

The per-interface decomposition and the two CDR metrics point the same way. MHC–peptide interfaces are modelled at DockQ $\approx$ 0.94 by all four methods; CDR3 loops are built to within 1.6–2.3 Å heavy-atom RMSD of the native by all four after framework alignment, with a spread across methods of half an ångström. Docking accuracy, by contrast, spans tens of ångströms, is right-skewed with catastrophic outliers, and accounts for essentially all of the difference between methods and between complexes. This reproduces the decomposition Shi et al. reported [27] on a larger and more recent set, and agrees with Lu et al.'s conclusion that the bottleneck is not the peptide in the groove but the TCR docking orientation [25], and it argues that further progress on TCR–pMHC modelling will come from the placement problem, not from loop refinement. That the placement should be the hard part is not obvious in advance: TCR docking polarity is conserved across the solved structures and is thought to be partly germline-encoded [5,6], so the geometry a model has to reproduce is closer to a canonical arrangement than to an unconstrained docking problem.

4.3 Cross-architecture agreement as a reference-free quality signal

The Cα RMSD between an AlphaFold 3 and an ESMFold2 prediction of the same sequence predicted the worse of their two DockQ scores at $\rho = -0.884$, against +0.60 to +0.64 for the models' own ipTM values, ~0.68 for pDockQ, and ~0.67–0.73 for learned quality scores. This is not a subtle effect and it is cheap: one extra prediction from a method that runs in five seconds. It also has a mechanistic reading. Two models that share a generative head but nothing on the conditioning side have largely independent failure modes for the docking problem, so their agreement is evidence in a way that a single model's self-reported confidence — which, as 7SG1 shows, can be high on a wrong answer — is not. A practical consequence follows: for TCR–pMHC modelling campaigns, running both architectures, using their agreement as a triage score, and taking the union of their predictions raises the medium-or-better rate on this benchmark from 85% (AlphaFold 3 alone) to 94%.

4.4 Sampler settings and computational cost

AlphaFold 3's denoising trajectory has converged to within 1 Å of its own answer by step 158 of 200, and running with 160 steps and a single diffusion sample returned the answer in 55% of the GPU time with no detectable accuracy cost over 30 complexes. The scope of that result should be emphasised: n = 30, one GPU, one seed per complex, and one narrow size regime (397–416 residues, all in AlphaFold 3's 512-token bucket). Larger complexes shift the trunk/diffusion split — the trunk scales roughly quadratically in the pair track while diffusion scales with atom count — so the 1.9× ceiling moves. Within this regime, though, the finding is straightforward: for high-throughput TCR–pMHC screening the default settings buy almost nothing, and the compute is better spent on a second architecture, which Section 3.9 shows is worth considerably more than a fifth diffusion sample.

4.5 Implications for benchmark design

Two methodological points generalise beyond this set. First, native construction is not a solved preprocessing step. Forty-six of 136 entries are built from files holding several complete complexes, and the deposited asymmetric unit is often not a complex; six natives required a biological-assembly fetch or a periodic-boundary repair, and one (9J4S) had no physical TCR–pMHC interface in any author-defined assembly, its apparent interface being a lattice contact between two copies. A contact-count check cannot catch this — only requiring all chains to come from a single author-defined assembly does. A benchmark that scores such an entry is scoring noise. Second, redundancy filtering on TCR identity is less informative than it appears for this class of complex: germline V-gene framework dominates a ~110-residue variable domain, so 119 of 126 targets sit above 70% identity to something pre-cutoff, and the $\leq$ 90% subset is 14 mostly non-human or atypical TCRs rather than a representative hard set. The composite identity to the closest *complete* pre-cutoff complex separates the benchmark far better and at usable sample sizes (46 targets have no complete pre-cutoff match at all). We report both, and we report explicitly that the AlphaFold 3–over–Protenix result does not survive a 95% TCR-identity filter, which is the kind of statement benchmark papers should make routinely.

4.6 Limitations

The class II subset (n = 15) is too small to resolve class effects, and TCRmodel2 was run on class I only because it accepts a fixed class II peptide length. Nine entries were excluded for non-standard peptide chemistry or covalent tethering, both of which are biologically interesting categories that sequence-input prediction currently cannot address, so their exclusion reflects a limitation of the methods as much as of the benchmark. "Training set" is a proxy throughout — neither AlphaFold 3's nor Protenix's actual training list is public, and we use PDB availability on or before 30 September 2021 as the closest reproducible stand-in, which cannot account for sequence databases or non-TCR structures the models also saw. The reference set for that analysis is only as complete as TCR3d $\cup$ STCRDab [15–17]. AlphaFold 3 and Protenix were run through public servers, so their exact data pipelines are outside our control; ESMFold2 was likewise run through the ESM SDK API, and the trajectory analysis of Section 3.10 used the local `biohub/ESMFold2` checkpoint rather than the API model, so its per-frame DockQ values are not interchangeable with the benchmark's. Finally, the sampler-economy result is a single-seed, single-GPU, 30-complex experiment in one size regime, as stated above.

5 Conclusions

On 126 TCR–pMHC complexes released after the shared 30 September 2021 training cutoff, a single-sequence protein-language-model diffusion predictor matched AlphaFold 3 and beat an AlphaFold 3 reproduction, with no MSA and no structural template. That reverses the standing conclusion of the recent benchmarks of this class of complex [25,26], and it does so without a compensating caveat: ESMFold2 was simultaneously indistinguishable from AlphaFold 3 on median accuracy, ahead of it on the fraction of medium-or-better models, better calibrated in its own confidence, and roughly two and a half times faster end to end. The evolutionary signal carried by a paired alignment, long treated as the ingredient that makes TCR–pMHC modelling work, is on this evidence recoverable from a sufficiently large language model — a change in what these predictors require, not only in how well they perform. Both models, and a TCR-specialised AlphaFold 2 derivative, build the pMHC and the CDR3 loops to comparable accuracy and differ almost entirely in where they place the TCR. The disagreement between the two architectures is anatomically confined to that placement, and predicts accuracy better than any of the confidence scores evaluated here, so that a two-architecture ensemble is both more accurate (94% versus 85% medium-or-better) and self-triaging. AlphaFold 3's denoising trajectory converges by step 158 of 200, and reducing the sampler to 160 steps and one diffusion sample returns the same answers in 55% of the GPU time. Taken together, these observations favour running two architectures at reduced sampler settings over a single architecture at default settings, and using the agreement between them to identify which predictions can be relied upon.

Data and code availability

The benchmark inputs (136 truncated FASTAs, chain manifests, review records), the rebuilt experimental references and their provenance manifest, all predictions, and the complete scoring, analysis and figure code are contained in the accompanying repository. Per-interface DockQ results are provided in long format (one row per interface plus Global DockQ), TCR CDR RMSD results per prediction with full coverage and quality-control columns, CDR lDDT and pLDDT in long format (one row per prediction and region, both contact sets), the sampler-economy tracks with per-complex timings and paired differences. The pre-cutoff audit of Section 3.8 is provided in full — per-target rules, identities, clusters and recommendations — together with the scripts that regenerate it. All analyses are reproducible from the recorded commands; software versions and model checkpoint revisions and hashes are recorded in the accompanying manifests.

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